Forced Internationalization Penalty: Firm Performance in Pacific Small Island Developing States

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Abstract

The conventional inverted-U between internationalization intensity and firm performance rests on three structural prerequisites: a viable domestic market, manageable trade costs, and functional institutions. When all three are absent at once, as in Pacific Small Island Developing States (SIDS), the non-linear pattern collapses into a monotone negative relationship, which we term the Forced Internationalization Penalty (FIP): not a transient phase-1 learning cost, but a persistent structural condition arising from the compulsion to export despite systemic disadvantage. Using World Bank Enterprise Survey data from 1,469 firms across nine Pacific SIDS (2009–2025), internationalization intensity is robustly negatively associated with labour productivity ($\beta = -0.404$, $p = .032$, country-and-year fixed effects) with no quadratic curvature. Technological capability and digital adoption, significant moderators in mainland Asia, are null here. These findings reframe the inverted-U as conditional rather than universal and identify boundary conditions for institutional moderation theory.

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1. Introduction

The relationship between internationalization and firm performance (I-P) is among the most contested empirical regularities in international business (IB) research. Meta-analyses consistently report moderate positive effects on average (Kirca et al., 2011; Bausch & Krist, 2007). Yet the variance across studies is enormous and heterogeneous, which is itself an invitation to examine the *conditions* under which internationalization helps or harms performance (Verbeke & Brugman, 2018).

The dominant theoretical resolution posits an inverted-U functional form (Lu & Beamish, 2004): internationalization first improves performance through economies of scale, learning, and market diversification; then, beyond a threshold, coordination costs and organizational complexity erode these gains. Contractor, Kundu, and Hsu (2003) extended this to a three-stage S-curve, in which even an initial downward phase-1 slope is transient and eventually gives way to net benefits. Both frameworks share a foundational assumption. The institutional environment, while it may moderate the *intensity* of the I-P relationship, does not alter its fundamental shape or sign.

This study challenges that assumption. We argue that certain structural conditions, characteristic of Small Island Developing States (SIDS), eliminate the prerequisites on which the inverted-U logic depends, and that they replace the expected non-linear pattern with a monotone negative one. We term this phenomenon the **Forced Internationalization Penalty (FIP)**: the persistent negative effect of internationalization intensity on firm performance in contexts where three structural prerequisites are absent at once.

We position FIP not as a peripheral anomaly but as the opposite *pole* of the I-P relationship, the theoretical counterpart against which conventional theory is properly understood. The standard inverted-U describes *internationalization-to-scale*: in strong-institution economies, exporting is a profit-optimizing choice, and the descending limb arrives only at a very late turning point because a viable home base, manageable trade costs, and functional institutions allow firms to appropriate scale and learning gains. FIP describes the inverse logic, *internationalization-to-survive*: when those same prerequisites are absent, firms are forced to export despite logistics costs so severe that they erode all profit, and the inverted-U collapses into a monotone negative line from the origin. The two regimes are not different points on one curve but two ends of a single theoretical continuum, with conventional theory occupying one pole and FIP the other. Far from a footnote, the FIP regime is the theoretically most revealing case. It is the analogue of a singularity, the limiting point at which the familiar laws cease to apply and, precisely in their breakdown, expose the structural prerequisites the inverted-U silently assumes but never names. Studying where the curve collapses tells us what the curve was built on.

SIDS offer a theoretically extreme yet empirically underserved context for I-P research. These economies are marked by microscopic domestic markets (Tonga: 104,000 inhabitants; Kiribati: 121,000), by geographic isolation so severe that it amplifies trade costs by 30–210% (Winters & Martins, 2004), and by comprehensive institutional voids that limit firms' ability to appropriate returns from international activities (Briguglio, 1995; Khanna & Palepu, 2010). Recent systematic tracking across 121,249 development data points confirms that these conditions have deepened rather than abated. Progress in 2020–2025 is the slowest on record across 15 of 26 tracked benchmarks, with SIDS and Sub-Saharan Africa disproportionately affected (Pirlea et al., 2026). An independent multi-indicator analysis of six development dimensions, namely poverty, life expectancy, schooling, gender equality, climate, and electricity, finds the world developing at its slowest pace in 75 years; 54 economies remain more than a century from reaching developed-country standards on current trajectories, and SIDS are disproportionately represented in this cohort (Mahler, Serajuddin, Wadhwa, & Yonzan, 2026). One point matters above the rest. Firms in SIDS do not *choose* to internationalize as a growth strategy; for many, exporting is a structural imperative dictated by the impossibility of achieving minimum efficient scale at home. This compulsion, which we describe as survival-based internationalization, fundamentally alters the logic of the I-P relationship.

This study makes three contributions to the IB literature. First, and centrally, it introduces and theoretically defines FIP as a *general* boundary condition for institutional moderation theory, not a niche curiosity confined to remote islands. By identifying the conditions under which the inverted-U inverts, FIP demonstrates that the inverted-U is itself conditional rather than universal: it is the special case that obtains when three structural prerequisites happen to be satisfied, and FIP is the equally lawful regime that obtains when they are not. SIDS are the empirical site where this becomes visible, but the contribution is to the architecture of I-P theory writ large. We distinguish FIP from adjacent concepts that include phase-1 costs in three-stage theory (Contractor et al., 2003), Liability of Foreignness (Zaheer, 1995), and necessity entrepreneurship (Sarasvathy et al., 2014). Second, it provides empirical evidence from 1,469 firms across nine Pacific SIDS economies using World Bank Enterprise Survey (WBES) data, the most comprehensive firm-level I-P analysis in Pacific SIDS to date. Third, it documents that firm-level capabilities, namely technological capability and digital adoption, do not appear to override FIP. This last result shifts the theoretical conversation from "institutional moderation of the I-P relationship" (Marano, Tallman, & Teece, 2016) toward "institutional determination of the sign of the I-P relationship."

The findings carry direct policy implications under the Antigua and Barbuda Agenda for SIDS (United Nations General Assembly, 2024). They suggest that strategies premised on export-led growth may be systematically misaligned with the structural realities of Pacific micro-economies.

2. Theoretical Background and Hypotheses

2.1 The Inverted-U Paradigm and Its Structural Prerequisites

The inverted-U hypothesis in I-P research derives from a cost-benefit logic: as firms expand into foreign markets, early positive effects (scale economies, knowledge acquisition, market diversification) initially dominate, but beyond a threshold, mounting organizational complexity, coordination costs, and the liability of foreignness reverse the performance trajectory (Lu & Beamish, 2004; Hitt, Hoskisson, & Kim, 1997). The turning point, typically estimated between 30% and 70% of foreign sales to total sales (FSTS), represents the point at which marginal costs equal the marginal benefits of further internationalization.

This logic rests, implicitly, on three structural prerequisites that are rarely stated but are routinely satisfied in mainstream IB research samples:

Prerequisite 1: A viable domestic market. The inverted-U logic assumes that firms have a functioning home base providing revenues, resources, and organizational learning before and during internationalization (Johanson & Vahlne, 1977; 2009). When the domestic market is non-viable, too small to support minimum efficient scale for even basic operations, firms are structurally compelled to export regardless of strategic readiness or capability. The "selection into exporting" assumption underlying most I-P models then breaks down.

Prerequisite 2: Manageable trade costs. The standard model assumes that, at least initially, export revenues exceed export costs, allowing a phase of net positive internationalization returns. In geographically isolated economies, trade costs may be so extreme that this condition is violated even for small export intensities. Winters and Martins (2004) documented trade cost premiums of 30–210% for remote small economies, largely independent of firm efficiency or strategic choice.

Prerequisite 3: Functional institutional support. Both institutional theory (Scott, 1995; North, 1990) and the institutional voids literature (Khanna & Palepu, 2010; Doh et al., 2017) treat the quality of market-supporting institutions as a moderator of I-P relationship intensity. This presupposes that institutions vary on a continuum where some level of functional support exists even at the low end. Comprehensive institutional voids tell a different story. Absent financial intermediaries, unreliable contract enforcement, and limited trade facilitation infrastructure do not merely weaken the inverted-U; they may eliminate the mechanism that generates positive returns from internationalization altogether.

2.2 Forced Internationalization Penalty: Theoretical Definition

We define the **Forced Internationalization Penalty (FIP)** as the monotone negative effect of internationalization intensity on firm performance in contexts where the three structural prerequisites above are simultaneously absent. FIP differs from related concepts in four respects:

From phase-1 costs (Contractor et al., 2003): Phase-1 costs are transient. They reflect initial setup and learning investments that yield net benefits once a turning point is crossed. FIP has no theoretical turning point within the observed range of FSTS values, because the constraint is structural (non-viable domestic market, extreme trade costs, institutional voids) rather than transitional.

From Liability of Foreignness (Zaheer, 1995): LoF denotes the additional costs incurred by foreign firms operating in host markets relative to domestic incumbents. FIP operates on the *home-market* side of the equation. It is the penalty SIDS firms incur when they export, not the penalty multinationals incur when entering SIDS markets.

From necessity entrepreneurship (Sarasvathy et al., 2014): Necessity entrepreneurs in developed contexts often achieve performance parity with opportunity-driven entrepreneurs given sufficient time and institutional support. FIP characterizes contexts where institutional support is systemically absent, so the necessity-driven internationalization path remains persistently disadvantaged.

From SIDS economic vulnerability (Briguglio, 1995): Briguglio's vulnerability index captures macroeconomic exposure to external shocks. FIP is a microeconomic, firm-level construct. It identifies the performance mechanism through which macro-structural vulnerability translates into firm-level outcomes.

2.3 The SIDS Context and Survival-Based Internationalization

Pacific SIDS economies exhibit all three violated prerequisites in extreme form. With populations ranging from 94,000 (Micronesia) to 10 million (Papua New Guinea, the largest in our sample), domestic markets cannot sustain manufacturing or service firms at minimum efficient scale. Read (2004) demonstrated that the smallest island economies have effectively no choice architecture for export abstention. Firms either export or exit. We term this pattern *survival-based internationalization*: not strategic expansion, but structural compulsion.

Under survival-based internationalization, FSTS, the standard measure of internationalization intensity, no longer indexes voluntary commitment to foreign markets. It proxies the *degree of domestic market inadequacy*. Firms with higher FSTS are not necessarily those with superior international capabilities; they may simply face more acute domestic market constraints. This reinterpretation of FSTS in the SIDS context generates a novel empirical prediction:

Hypothesis 1 (FIP hypothesis): In Pacific SIDS, internationalization intensity is negatively associated with firm performance with no significant quadratic turning point, indicating a Forced Internationalization Penalty rather than an inverted-U relationship.

2.4 The Role of Firm-Level Capabilities

In mainland Asian and other institutional contexts, firm-level capabilities moderate the I-P relationship. Technological capability (Lall, 1992; Goedhuys & Sleuwaegen, 2013) and digital adoption (Teece, 2007) have been shown to shift the I-P curve upward, enabling firms to reach better performance at any given level of internationalization, and to alter the turning point. The theoretical mechanism is that capabilities reduce the organizational complexity costs on the descending portion of the inverted-U.

In the FIP context, this moderation mechanism may be compromised. If the negative I-P relationship stems from structural conditions (non-viable domestic market, extreme trade costs, institutional voids) rather than from organizational complexity, then firm-level capabilities that address organizational complexity will not redirect the relationship. A SIDS firm with superior technological capabilities still faces trade cost amplifications of 30–210%. A SIDS firm with a website still faces the absence of functional financial intermediaries for trade finance. The structural mechanism that generates FIP operates at a level firm-level capabilities cannot easily offset. Two labels make this concrete. First, the negative slope is an *Isolation Penalty*, a transaction-cost-of-isolation that geographic remoteness imposes on every cross-border transaction (the 30–210% trade-cost amplification), which a compelled exporter must bear permanently rather than amortize away. Second, a digital shield presupposes a minimal payment-and-logistics substrate to function; under the *extreme institutional void* characteristic of SIDS, that substrate is largely absent, so digital capability cannot substitute for the missing institutions and physical infrastructure.

Hypothesis 2 (Capability null hypothesis): In Pacific SIDS, technological capability and digital adoption do not significantly moderate the internationalization–performance relationship.

Note: H2 is formulated as a null-expectation hypothesis based on the FIP theoretical mechanism, not a power argument. This framing follows the methodological guidance of Haans, Pieters, and He (2016) on hypothesizing null results for theoretically motivated reasons.

3. Data and Methods

3.1 Data Source

We use pooled data from the World Bank Enterprise Survey (WBES), drawing waves conducted in Pacific SIDS economies between 2009 and 2025. WBES employs stratified random sampling with probability-proportional-to-size design across firm-size strata (5–19, 20–99, 100+ employees) and sectors (manufacturing, services, retail). We include all firm-year observations from economies meeting the SIDS classification: Fiji, Kiribati, Papua New Guinea, Samoa, Solomon Islands, Timor-Leste, Tonga, Vanuatu, and Comoros (nine economies). The analysis sample, observations with non-missing values for labor productivity, export intensity, and firm age, comprises **1,469 firms** across multiple country-year waves.

3.2 Measures

Dependent variable. Log labor productivity (ln_LP) is constructed as the natural logarithm of sales revenue per permanent worker, expressed in constant purchasing-power-parity-adjusted USD: ln(d2/l1). This measure has been used extensively in enterprise-level productivity studies (Goedhuys & Sleuwaegen, 2013; World Bank, 2020). Of the 1,469 analysis observations, 187 (12.7%) are exporters with FSTS > 0.

Internationalization intensity. Following the methodology of Haans et al. (2016) and Contractor et al. (2003), we construct FSTS (Foreign Sales to Total Sales) from WBES item d3c (percentage of sales exported), divided by 100. For the main models, FSTS is mean-centered at the wave level: FSTS_c = FSTS − mean_wave(FSTS), yielding a mean FSTS of 0.060 (SD unreported due to small exporter subsample). Mean-centering suppresses cross-wave level differences while preserving within-wave variation, following standard practice for studies examining FSTS curvature (Haans et al., 2016).

Technological capability index (TCI_z). Following Lall (1992) and Goedhuys and Sleuwaegen (2013), TCI is constructed from two WBES items: b8 (internationally recognized quality certification, binary) and e6 (licensing of foreign technology, binary). TCI_z is the z-standardized mean of these two indicators.

Digital adoption index (DAI_z). DAI is measured using WBES item c22b (firm has an operational website, binary). This is a Tier-1 proxy capturing *foundational website adoption*, the first step of digital presence. It is appropriate for the SIDS context where advanced digital infrastructure is limited, only 5% of low-income country populations had basic digital skills in 2023, and high-income economies account for 77% of global data center capacity (Mahler, Wang, & Weber, 2026). DAI_z is the z-standardized value. We do not claim to measure dynamic digital capability, which would require multi-tiered indicators unavailable in WBES for this sample.

Control variables. Firm size is the natural logarithm of the number of permanent workers (ln_size, from l1). Firm age is survey year minus year of establishment (b5). Foreign ownership (foreign_own_pct) is the percentage of firm equity held by foreign entities (b2b). These are standard controls in enterprise-level I-P research (Hitt et al., 1997; Lu & Beamish, 2004).

Table 0: Variable definitions and WBES construction.

Variable	WBES item(s)	Construction	Role in model
ln_LP	d2, l1	ln(annual sales PPP / permanent workers); PPP-deflated using World Bank ICP deflators by survey year	Dependent variable — log labor productivity (Goedhuys & Sleuwaegen, 2013)
FSTS	d3c	d3c / 100; proportion of total sales from exports	Internationalization intensity — raw (Haans et al., 2016)
FSTS_c	d3c	FSTS − mean_wave(FSTS); wave-level mean-centred	Internationalization intensity — centred (Contractor et al., 2003)
$FSTS\_c^2$	d3c	$FSTS\_c^2$; tests curvature (H1)	Quadratic term; $\beta_2 < 0$ if inverted-U, NS if FIP monotone (Lind & Mehlum, 2010)
TCI_z	b8, e6	z-std(mean(b8_binary, e6_binary)); quality cert + foreign tech licence	Technological capability — Tier-1 proxy; direct + moderation test (H2; Lall, 1992)

Variable	WBES item(s)	Construction	Role in model
DAI_z	c22b	z-std(c22b_binary); operational website presence — Tier-1 only	Digital adoption — foundational website proxy; NOT dynamic digital capability (Banalieva & Dhanaraj, 2019; H2)
ln_size	l1	ln(permanent employees)	Control — firm size (Hitt et al., 1997)
firm_age	b5	survey_year − b5	Control — firm age (Lu & Beamish, 2004)
foreign_own_pct	b2b	% equity held by foreign entities	Control — foreign ownership (Marano et al., 2016)
δ_c	a4a/a4b	Country fixed effects (9 SIDS economies)	Absorbs time-invariant country heterogeneity: institutional quality, remoteness, trade costs
λ_t	wave	Wave-year fixed effects (2009, 2015, 2016, 2018, 2019, 2022, 2023, 2024, 2025)	Absorbs common time trends

Notes. b8 and e6 are recoded from WBES response codes (1 = yes, 2 = no) to binary (1/0). c22b is recoded similarly. TCI_z and DAI_z are standardised across the full pooled sample. PPP conversion uses World Bank ICP deflators matched to survey year. FSTS_c is centred within each country-year wave so that turning-point estimates are not confounded by cross-wave composition shifts (Haans et al., 2016). DAI captures only Tier-1 static website presence; Tier-2 (e-payment) and Tier-3 (cloud/ERP) indicators are unavailable for these WBES waves.

3.3 Model Sequence

We estimate four model families to test H1 and H2:

M0 (controls baseline): ln_LP_it = α + γ_1 ln_size_it + γ_2 firm_age_it + γ_3 foreign_own_pct_it + δ_c + λ_t + ε_it

M1 (linear FSTS, FIP test): $\ln\_LP\_it = \alpha + \beta_1$ FSTS_c_it + γ·X_it + δ_c + λ_t + ε_it

M2 (quadratic FSTS, curvature test): $\ln\_LP\_it = \alpha + \beta_1$ FSTS_c_it + β_2 FSTS_c²_it + γ·X_it + δ_c + λ_t + ε_it

M3 (capabilities, moderation null test): $\ln\_LP\_it = \alpha + \beta_1$ FSTS_c_it + β_2 FSTS_c²_it + β_3 TCI_z_it + β_4 DAI_z_it + γ·X_it + δ_c + λ_t + ε_it

where δ_c denotes country fixed effects, λ_t wave-year fixed effects, and X the vector of control variables. Country fixed effects absorb time-invariant country-level heterogeneity (institutional quality, trade costs, geographic remoteness) that would otherwise confound the FSTS coefficient.

We also estimate two additional robustness specifications: M_yearFE (year fixed effects only, no country fixed effects) and M_bivariate (no controls), to assess the sensitivity of the FIP estimate to different modeling assumptions.

All regressions are estimated by OLS with heteroscedasticity-robust standard errors. The analysis is implemented in R; the full replication script, coefficient tables, and summary statistics are archived in `p8/replication/`.

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4. Results

4.1 Sample Description and Preliminary Evidence

The analysis sample spans 1,469 firms across nine Pacific SIDS economies. Table 1 reports country-level summary statistics. Exporter representation is thin across all economies (overall 12.7%), consistent with the SIDS structural constraint: most firms operate predominantly in the domestic market despite its limitations. Mean FSTS across the full sample is 0.060, reflecting the low-but-non-zero export engagement of SIDS firms. The country fixed effects absorb substantial variance: R^2 increases from 0.511 (year FE only) to 0.799 (country + year FE) when country indicators are added, confirming that 95% of explained productivity variance is between-country, a feature consistent with extreme between-country heterogeneity in institutional quality, geographic remoteness, and economic structure.

4.2 Main Results: Forced Internationalization Penalty (H1)

Table 2 reports the main regression results. **M1**, the linear FSTS specification with country and year fixed effects, yields:

β(FSTS_c) = **−0.404**, SE = 0.188, t = −2.152, p = **.032**

This coefficient is statistically significant at the conventional 5% level and substantively meaningful: a one-unit increase in FSTS (a move from no exports to full-export orientation) is associated with a 40.4% lower labor productivity. This supports **Hypothesis 1**: internationalization intensity is negatively associated with performance in Pacific SIDS.

The M0 baseline (controls only, no FSTS) yields an R^2 of 0.7992, and adding FSTS_c in M1 marginally increases R^2 to 0.8001, which indicates that, conditional on country and year fixed effects, FSTS explains limited additional variance but does so in a consistently negative direction.

Control variable results in M1 are consistent with established evidence: firm age is positively associated with productivity (β = +0.009, p = .052); foreign ownership shows a marginally negative association (β = −0.006, p = .056), perhaps reflecting thin foreign investment efficiency in SIDS contexts; firm size is positive but non-significant (β = +0.082, p = .242).

4.3 Monotone Negative Form: No Curvature (H1, quadratic null)

M2 introduces the quadratic term FSTS_c² to test whether the negative I-P relationship is eventually reversed, i.e., whether there is any inverted-U pattern with the positive portion below our data range.

The results are: β(FSTS_c) = −0.925, SE = 0.828, p = .265 (n.s.) β(FSTS_c²) = +0.649, SE = 0.980, p = .508 (n.s.)

Both the linear and quadratic terms lose significance when entered jointly, indicating high collinearity between the two terms in this low-FSTS sample (mean FSTS = 0.060). Importantly, the positive quadratic coefficient would imply a U-shape rather than inverted-U, a reversal of the conventional form that is itself consistent with FIP (the relationship is monotonically negative with no upward turning point within the observed range).

Following Haans et al. (2016), we evaluate the four necessary conditions for a curvature relationship. All four must hold for an inverted-U to be confirmed; here, all four *fail*, which is the expected signature of FIP:

- **C1** (positive slope at low FSTS): β(FSTS_c) = −0.925 (p = .265), **not significant and negative in sign**; no ascending limb at low FSTS values
- **C2** (negative curvature $\beta_2 < 0$): β(FSTS_c²) = +0.649 (p = .508), **not significant; positive sign indicates U-shape, not inverted-U**
- **C3** (turning point within data range): TP* = −(−0.925) / (2 × 0.649) = 0.712, **implied turning point lies outside the observed sample maximum (FSTS_max ≈ 0.80, but neither β_1 nor β_2 is statistically significant)**
- **C4** (Lind–Mehlum utest, Lind & Mehlum, 2010): utest p = .941, **fails to reject the null of monotone negative relationship; no inverted-U detected**

The pattern is **monotone negative**: consistent with H1 and the FIP structural mechanism. There is no turning point within the observed FSTS range (0–1).

4.4 Robustness

We examine three additional specifications to assess the robustness of the FIP result.

Specification M_yearFE (year fixed effects only): Removing country fixed effects, which might absorb too much structural variation, yields a substantially larger negative effect:

β(FSTS) = **−1.236**, SE = 0.269, t = −4.594, p = **<.001**

This larger coefficient reflects the additional between-country variation in the FSTS–performance relationship. The FIP result strengthens considerably, confirming that country-level structural disadvantages compound firm-level FIP.

Specification M_bivariate: A bivariate regression of ln_LP on FSTS (no controls) yields:

β(FSTS) = **−1.596**, SE = 0.263, t = −6.062, p = **<.001**

The monotone negative pattern persists without any conditioning on firm characteristics, further confirming the robustness of FIP.

Exporters-only subsample (N = 187): Among the 187 active exporters (FSTS > 0), the negative effect is:

β(FSTS_c) = **−0.901**, SE = 0.398, p = **.027**

This result is critical: even among firms that have successfully engaged in exporting, higher export intensity is associated with lower productivity. This rules out a simple selection explanation (e.g., that only low-productivity firms are excluded from exporting, driving the pooled negative correlation). Among actual exporters, intensifying exports further worsens performance, which is the hallmark of the FIP mechanism rather than of adverse selection.

4.5 Capability Null Results (H2)

M3 introduces TCI_z and DAI_z as additional regressors to test H2. Neither variable achieves statistical significance:

β(TCI_z) = +0.058, SE = 0.085, $p = .495$ (n.s.) β(DAI_z) = +0.062, SE = 0.073, $p = .402$ (n.s.)

The FIP coefficient (FSTS_c) remains negative but loses significance in M3 ($\beta = -0.974$, $p = .252$), likely reflecting the substantially reduced sample (N = 526 for M3, versus N = 1,469 for M1, due to missing capability variables). The direction of the FSTS_c coefficient remains negative and numerically larger in M3, consistent with FIP; the significance loss is attributable to power reduction from the subsample restriction.

These results are consistent with H2 (non-moderation): technological capability (TCI_z: $\beta = +0.058$, SE = 0.085, $p = .495$) and digital adoption (DAI_z: $\beta = +0.062$, SE = 0.073, $p = .402$) do not statistically significantly moderate the I-P relationship in Pacific SIDS, consistent with the FIP mechanism overwhelming capability effects in structurally constrained SIDS economies. The positive point estimates for TCI_z and DAI_z, while small and non-significant, are consistent with the interpretation that capabilities raise productivity levels in general (as found in mainland Asian contexts: Goedhuys & Sleuwaegen, 2013; Teece, 2007) but cannot modify the structural negativity of the FSTS–performance relationship that FIP describes.

4.6 Summary

Table 2 summarizes the full coefficient matrix. The FIP pattern is confirmed: (i) a statistically significant negative linear FSTS–performance relationship (M1: $\beta = -0.404$, $p = .032$); (ii) no significant quadratic curvature or turning point (M2: both FSTS_c and FSTS_c^2 non-significant); (iii) robustly negative across four specifications (M_yearFE: $\beta = -1.236$, $p < .001$; M_bivariate: $\beta = -1.596$, $p < .001$; exporters-only: $\beta = -0.901$, $p = .027$); and (iv) null capability moderation (M3: TCI_z $p = .495$; DAI_z $p = .402$).

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5. Discussion

5.1 FIP as Boundary Condition for the Inverted-U Paradigm

Our central finding is that the inverted-U I-P relationship does not hold in Pacific SIDS. Instead of a curvilinear pattern with a positive phase, the data show a monotone negative relationship. It persists across four modeling specifications, and it is stronger among actual exporters than in the full sample. This is inconsistent with the inverted-U prediction (Lu & Beamish, 2004) and with the three-stage S-curve (Contractor et al., 2003), both of which predict at least some range of positive internationalization–performance association.

The FIP concept explains this reversal through the logic of violated structural prerequisites. When no viable domestic market exists, FSTS is not a strategic choice variable but a survival necessity, and firms cannot accumulate the home-base resources and organizational capabilities that fund and sustain profitable internationalization. When trade costs amplify the "cost of doing business" of export revenues by 30–210% (Winters & Martins, 2004), the marginal economics of exporting are negative from the outset. And when institutional voids are comprehensive, with absent trade finance intermediaries, unreliable contract enforcement, and limited standards infrastructure, firms cannot appropriate the learning and scale benefits that internationalization theory promises.

This reframes the inverted-U from a universal empirical regularity into a *conditional* relationship: it holds when and because the structural prerequisites are satisfied. The existing literature's finding that institutional quality *moderates* the I-P relationship (Marano et al., 2016) is then a special case of a more general theoretical relationship. When institutional conditions cross a structural threshold from "weaker moderation" to "violated prerequisites," the sign of the I-P relationship reverses, not merely its magnitude.

Read this way, conventional theory and FIP are the two poles of a single continuum rather than a rule and its exception. At the strong-institution pole, internationalization is internationalization-to-scale: a firm such as a Singaporean exporter operates on the long ascending limb where exporting optimizes profit margins, and the turning point arrives very late because the home base, trade costs, and institutions all permit scale and learning gains to be appropriated. At the opposite pole, internationalization is internationalization-to-survive: SIDS firms are forced to export despite trade-cost amplifications that erode all profit, and the curve collapses into a monotone negative line. The value of the FIP pole is not that it is rare but that it is diagnostic. It is the theoretically most revealing case precisely because it is the singularity at which the inverted-U breaks down, and where the breakdown lays bare the structural prerequisites the standard model assumes but leaves unstated. The conventional curve never had to declare its dependence on a viable home market, on bearable logistics, and on functioning institutions, because in mainstream samples those conditions are always present; only at the FIP pole, where they vanish together, does that hidden dependence become observable. FIP therefore contributes a general boundary condition to institutional moderation theory: it specifies the limit at which moderation of degree gives way to determination of sign.

5.2 Theoretical Positioning: Moving from Degree to Kind

FIP represents a qualitative shift in how we conceptualize institutional effects on I-P relationships. Most prior theorizing treats institutional effects as continuous modifiers of the inverted-U's shape and location: higher institutional quality shifts the optimum FSTS upward, broadens the positive range, or raises the peak performance level (Marano et al., 2016; Hitt et al., 1997). This is a change in *degree*.

Our findings point to a change in *kind*. At the extreme end of the institutional spectrum, where all three structural prerequisites are absent at once, the I-P functional form is not a modified inverted-U but a qualitatively different pattern, namely a monotone negative one. This extends the theoretical vocabulary of IB research beyond "intensity moderation" toward "existence moderation" of the I-P relationship.

The finding connects to the broader question of boundary conditions in IB theory (Verbeke & Brugman, 2018; Hennart, 2007). If mainstream I-P theories are derived from samples of economies where the structural prerequisites are broadly satisfied, those theories may be systematically inapplicable to the SIDS context. They would not merely need adjustment; they would represent a qualitatively different theoretical regime. FIP offers a theoretically grounded label for that alternative regime.

5.3 The Capability Null: Institutional Determination vs. Moderation

The null results for TCI_z and DAI_z in M3 add a second dimension to the FIP theory. In mainland Asian contexts, including Vietnam, Singapore, China, and multi-country samples, technological capability and digital adoption are statistically significant moderators of the I-P relationship (Goedhuys & Sleuwaegen, 2013; Teece, 2007). Why, then, are they null in SIDS?

The FIP mechanism provides the answer. Capabilities moderate *organizational complexity costs* on the descending portion of the inverted-U, but FIP operates at a structural, pre-organizational level. A SIDS firm with ISO 9001 certification (high TCI) still faces the same 30–210% trade cost amplifications; a SIDS firm with a functioning website (DAI = 1) still faces absent trade finance institutions. Capabilities that act as powerful moderators within the structural prerequisites framework become ineffective once those prerequisites do not hold.

This is consistent with the resource-based view's emphasis on context-dependency (Barney, 1991): resources generate value only within institutional environments that support appropriability. Teece's (2007) dynamic capabilities framework similarly assumes functioning market mechanisms. In their absence, dynamic capabilities may be "trapped" inside the firm, unable to translate into performance advantages through international market transactions.

The positive though non-significant point estimates for TCI_z and DAI_z suggest that capabilities do raise general productivity levels, consistent with their role in the broader literature, yet they cannot modify the negative directionality of the FSTS–performance relationship in the SIDS structural context. This nuance, that capabilities help the level and not the slope, may be a productive direction for future SIDS research.

5.4 Policy Implications: The Antigua and Barbuda Agenda

Our findings have direct implications for SIDS development policy. The Antigua and Barbuda Agenda for SIDS (United Nations General Assembly, 2024), adopted at the Fourth International Conference on SIDS, frames export promotion and trade facilitation as central pillars of SIDS development strategy. This framing implicitly assumes that increasing FSTS will improve firm performance and thereby stimulate economic growth.

Our results caution against this assumption. If FIP is a structural feature of Pacific SIDS firm economies, then policies that push SIDS firms toward higher export intensity, without addressing the violated structural prerequisites in parallel, may worsen firm performance at the micro level even while they improve macroeconomic trade balances. The relevant policy question becomes: which interventions can address the three structural prerequisites that generate FIP?

On **Prerequisite 1** (viable domestic market): regional integration initiatives (Pacific Agreement on Closer Economic Relations Plus, PACER Plus) that create effective market access to neighboring SIDS and to Australia and New Zealand could expand the effective domestic market beyond national borders, reducing the structural compulsion to export to distant high-cost markets.

On **Prerequisite 2** (manageable trade costs): transport cost reduction through regional maritime connectivity programs and, where feasible, air cargo subsidies for high-value, low-weight exports (agricultural products, crafts, services) can reduce the trade cost amplification that drives FIP.

On **Prerequisite 3** (functional institutions): targeted institutional development focused on trade facilitation infrastructure, such as standards bodies, trade finance intermediaries, and export promotion agencies with genuine capacity, may address the FIP mechanism more directly than generic governance reform. Goedhuys and Sleuwaegen's (2013) evidence on international certification is instructive here. Their finding that ISO certification improves firm productivity is consistent with the null TCI result reported above if certification's productivity benefit is conditional on institutions that support standards enforcement and customer recognition, the very conditions absent in the SIDS context.

5.5 Limitations

Several limitations bound the interpretation of our findings.

First, **cross-sectional pooled structure**: although we pool across multiple country-year waves (2009–2025), few SIDS economies have more than one WBES wave (see Table 1), which limits panel identification of causal effects. The FIP estimate is identified primarily from cross-sectional variation in FSTS within country-year cells. Future work using panel data, where available for Fiji and Papua New Guinea, both of which have two waves, could strengthen causal identification.

Second, **FSTS measurement in thin-export contexts**: with a mean FSTS of 0.060 and only 12.7% exporters, the FSTS distribution is highly right-skewed. Coefficient estimates in M2 may be sensitive to a handful of high-FSTS outliers, so the non-significant quadratic results should be read with caution. Future work could employ quantile regression or winsorization to assess distributional sensitivity.

Third, **DAI as Tier-1 proxy only**: our digital adoption measure captures only website presence, a foundational and static indicator. This is appropriate for the SIDS context given the limited availability of more advanced digital infrastructure indicators in WBES, but it does mean we cannot assess whether more sophisticated digital capabilities (e-commerce, mobile payments, cloud computing) might moderate FIP differently. We explicitly do not claim to measure dynamic digital capability.

Fourth, **generalizability beyond Pacific SIDS**: FIP is theorized to arise when three structural prerequisites are absent at once. Other extreme contexts, such as Saharan micro-states, remote highland economies, and post-conflict settings, may exhibit similar patterns. Caribbean SIDS, which are not included in our sample, represent a natural extension context.

6. Conclusions

This study introduced the Forced Internationalization Penalty (FIP), the monotone negative effect of internationalization intensity on firm performance when three structural prerequisites of the inverted-U relationship are absent at once: a viable domestic market, manageable trade costs, and functional institutional support. Using WBES data from 1,469 firms across nine Pacific SIDS economies, we find robust support for FIP across four empirical specifications ($\beta = -0.404$, $p = .032$ in the primary country-and-year fixed-effects model; $\beta = -1.236$, $p < .001$ without country fixed effects; $\beta = -0.901$, $p = .027$ in the exporters-only subsample). The quadratic term is not significant, which confirms the monotone negative form with no observable turning point. Technological capability and digital adoption, both significant in mainland Asian contexts, are null in the SIDS environment, consistent with the FIP theoretical mechanism.

These findings contribute to IB theory in two ways. First, they help reorient the discourse on institutional effects from "institutional moderation of I-P intensity" toward "institutional determination of I-P sign," which establishes the inverted-U as conditional on structural prerequisites rather than universal. FIP and conventional theory are best understood as the two poles of one continuum: internationalization-to-scale, where exporting optimizes margins and the turning point is very late, and internationalization-to-survive, where firms are compelled to export despite profit-eroding logistics costs and the curve collapses to a monotone negative line. The SIDS case is valuable not because it is exotic but because it is the singularity at which the standard model breaks down and thereby reveals the prerequisites it silently assumes; FIP names that limiting regime and supplies a general boundary condition for institutional moderation theory. Second, they extend the application of "forced internationalization" beyond the SME literature into the performance-outcome domain, providing a theoretically defined and empirically grounded construct, FIP, that future research can build upon.

For the world development policy agenda, FIP suggests that export promotion strategies in SIDS require complementary structural interventions. These should address trade costs, domestic market viability through regional integration, and targeted institutional development, rather than the intensification of export activity on its own.

References

Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management, 17*(1), 99–120.

Bausch, A., & Krist, M. (2007). The effect of context-related moderators on the internationalization-performance relationship. *Management International Review, 47*(3), 319–347.

Briguglio, L. (1995). Small island developing states and their economic vulnerabilities. *World Development, 23*(9), 1615–1632.

Briguglio, L., Cordina, G., Farrugia, N., & Vella, S. (2009). Economic vulnerability and resilience: Concepts and measurements. *Oxford Development Studies, 37*(3), 229–247.

Contractor, F. J., Kundu, S. K., & Hsu, C.-C. (2003). A three-stage theory of international expansion: The link between multinationality and performance in the service sector. *Journal of International Business Studies, 34*(1), 5–18.

Diamantopoulos, A., & Winklhofer, H. M. (2001). Index construction with formative indicators: An alternative to scale development. *Journal of Marketing Research, 38*(2), 269–277.

Doh, J. P., Rodrigues, S., Saka-Helmhout, A., & Makhija, M. (2017). International business responses to institutional voids. *Journal of International Business Studies, 48*(3), 293–307.

Goedhuys, M., & Sleuwaegen, L. (2013). The impact of international standards certification on the performance of firms in less developed countries. *World Development, 46*, 87–101.

Haans, R. F. J., Pieters, C., & He, Z.-L. (2016). Thinking about U: Theorizing and testing U- and inverted U-shaped relationships in strategy research. *Strategic Management Journal, 37*(7), 1177–1195. https://doi.org/10.1002/smj.2399

Hennart, J.-F. (2007). The theoretical rationale for a multinationality-performance relationship. *Management International Review, 47*(3), 423–452.

Hitt, M. A., Hoskisson, R. E., & Kim, H. (1997). International diversification: Effects on innovation and firm performance in product-diversified firms. *Academy of Management Journal, 40*(4), 767–798.

Jarvis, C. B., MacKenzie, S. B., & Podsakoff, P. M. (2003). A critical review of construct indicators and measurement model misspecification in marketing and consumer research. *Journal of Consumer Research, 30*(2), 199–218.

Johanson, J., & Vahlne, J.-E. (1977). The internationalization process of the firm. *Journal of International Business Studies, 8*(1), 23–32.

Johanson, J., & Vahlne, J.-E. (2009). The Uppsala internationalization process model revisited. *Journal of International Business Studies, 40*(9), 1411–1431.

Khanna, T., & Palepu, K. G. (2010). *Winning in emerging markets: A road map for strategy and execution*. Harvard Business Press.

Kirca, A. H., Hult, G. T. M., Roth, K., Cavusgil, S. T., Perry, M. Z., Akdeniz, M. B., Deligonul, S. Z., Mena, J. A., Pollitte, W. A., Hult, R. N., White, J. C., & White, R. C. (2011). Firm-specific assets, multinationality, and financial performance: A meta-analytic review and theoretical integration. *Academy of Management Journal, 54*(1), 47–72.

Lall, S. (1992). Technological capabilities and industrialization. *World Development, 20*(2), 165–186.

Lind, J. T., & Mehlum, H. (2010). With or without U? The appropriate test for a U-shaped relationship. *Oxford Bulletin of Economics and Statistics, 72*(1), 109–118.

Lu, J. W., & Beamish, P. W. (2004). International diversification and firm performance: The S-curve hypothesis. *Academy of Management Journal, 47*(4), 598–609.

Mahler, D., Wang, H., & Weber, M. (2026, May). Inequalities in use of and exposure to artificial intelligence. In *Atlas of Global Development 2026*. World Bank Data360. World Bank Group. https://data360.worldbank.org/en/int/atlas/artificial-intelligence

Marano, V., Tallman, S., & Teece, D. J. (2016). The context of innovation and the innovation of context. *Journal of Leadership & Organizational Studies, 23*(4), 421–428.

North, D. C. (1990). *Institutions, institutional change and economic performance*. Cambridge University Press.

Pirlea, A. F., Wadhwa, D., Mahler, D., Serajuddin, U., Welch, M., Thudt, A., & Lambrechts, M. (2026, May). Global progress. In *Atlas of Global Development 2026*. World Bank Data360. World Bank Group. https://data360.worldbank.org/en/int/atlas/global-progress

Read, R. (2004). The implications of increasing globalization and regionalism for the economic growth of small island states. *World Development, 32*(2), 365–378.

Sarasvathy, S. D., Kumar, K., York, J. G., & Bhagavatula, S. (2014). An effectual approach to international entrepreneurship: Overlaps, challenges, and provocative possibilities. *Entrepreneurship Theory and Practice, 38*(1), 71–93.

Scott, W. R. (1995). *Institutions and organizations*. Sage.

Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal, 28*(13), 1319–1350.

United Nations General Assembly. (2024). *The Antigua and Barbuda Agenda for Small Island Developing States: A renewed declaration for resilient prosperity* (A/RES/78/317). United Nations.

Vahlne, J.-E., & Johanson, J. (2020). The Uppsala model: Networks and microfoundations. *Journal of International Business Studies, 51*(1), 4–10.

Verbeke, A., & Brugman, M. (2018). Triple-testing the quality of multinationality–performance research: An internalization theory perspective. *Journal of International Business Studies, 49*(7), 801–822.

Winters, L. A., & Martins, P. M. G. (2004). When comparative advantage is not enough: Business costs in small remote economies. *World Trade Review, 3*(3), 347–383.

Zaheer, S. (1995). Overcoming the liability of foreignness. *Academy of Management Journal, 38*(2), 341–363.

Appendix A: Tables

Table 1. Country-Level Summary Statistics

Economy	N (firms)	Exporters	Exporter %	Mean FSTS	Waves
Fiji	165	46	27.9%	0.109	2009–2025
Kiribati	114	3	2.6%	0.011	2025
Papua New Guinea	141	20	14.2%	0.028	2015–2024
Samoa	131	24	18.3%	0.051	2023
Solomon Islands	107	10	9.3%	0.040	2025
Timor-Leste	393	53	13.5%	0.104	2009–2021
Tonga	144	13	9.0%	0.044	2024
Vanuatu	157	16	10.2%	0.037	2009–2023

Economy	N (firms)	Exporters	Exporter %	Mean FSTS	Waves
Comoros	117	2	1.7%	0.011	2025
Total	**1,469**	**187**	**12.7%**	**0.060**	2009–2025

Note: Country-level breakdowns are available in the replication archive (`p8/replication/`). Total N reflects analysis sample (non-missing ln_LP and FSTS). Wave coverage varies by country.

Table 2. Regression Results: Internationalization Intensity and Labor Productivity

	M0 Controls	M1 Linear	M2 Quadratic	M3 Capabilities	M_yearFE	M_bivariate
FSTS_c		−0.404*	−0.925	−0.974		
		(0.188)	(0.828)	(0.850)		
FSTS_c²			+0.649	+0.782		
			(0.980)	(1.001)		
FSTS					−1.236***	−1.596***
					(0.269)	(0.263)
TCI_z				+0.058		
				(0.085)		
DAI_z				+0.062		
				(0.073)		
ln_size	+0.080	+0.082	+0.084	+0.062	+0.247*	—
	(0.070)	(0.070)	(0.070)	(0.070)	(0.097)	
firm_age	+0.009*	+0.009.	+0.009.	+0.010*	+0.047***	—
	(0.005)	(0.005)	(0.005)	(0.005)	(0.009)	
foreign_own_pct	−0.006.	−0.006.	−0.006.	−0.006.	−0.006*	—
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	
Country FE	Yes	Yes	Yes	Yes	—	—
Year FE	Yes	Yes	Yes	Yes	Yes	—
N	1,469	1,469	1,469	526	1,469	1,469
R^2	0.799	0.800	0.800	—	0.512	—

*Note: Coefficients reported with heteroscedasticity-robust standard errors in parentheses. Country fixed effects absorbed in M0–M3. *** $p < .001$; ** $p < .01$; * $p < .05$; . p*

< .10.

Forced Internationalization Penalty (FIP): Confirmed, β(FSTS_c) = −0.404, p = .032 (M1).

Table 3. Robustness: Exporters-Only Subsample (N = 187)

Specification	β(FSTS_c)	SE	p-value	Direction
OLS, country+year FE, N=187	−0.901	0.398	.027*	Negative
M1 full sample	−0.404	0.188	.032*	Negative
M_yearFE full sample	−1.236	0.269	<.001***	Negative
M_bivariate	−1.596	0.263	<.001***	Negative

Note: Penalty confirmed in all specifications. Exporters-only result (β = −0.901) addresses adverse selection concern: among active exporters, intensifying exports further worsens performance.